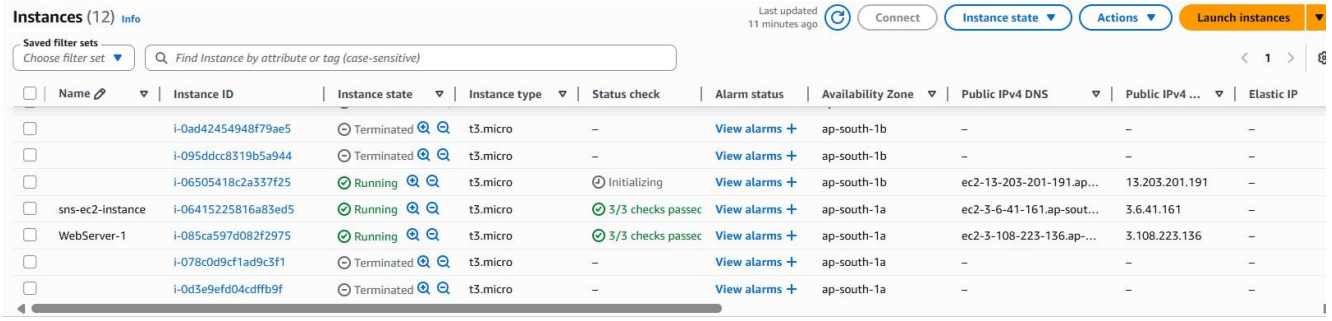
Name: Dhanashri Ishwar Wadile

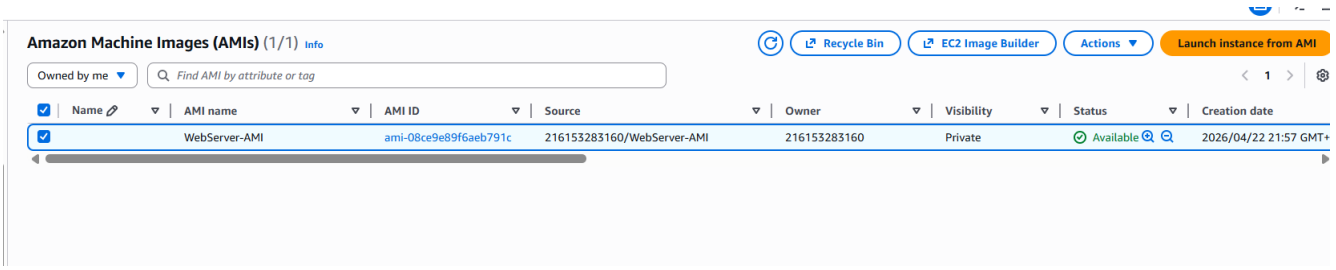
Project :

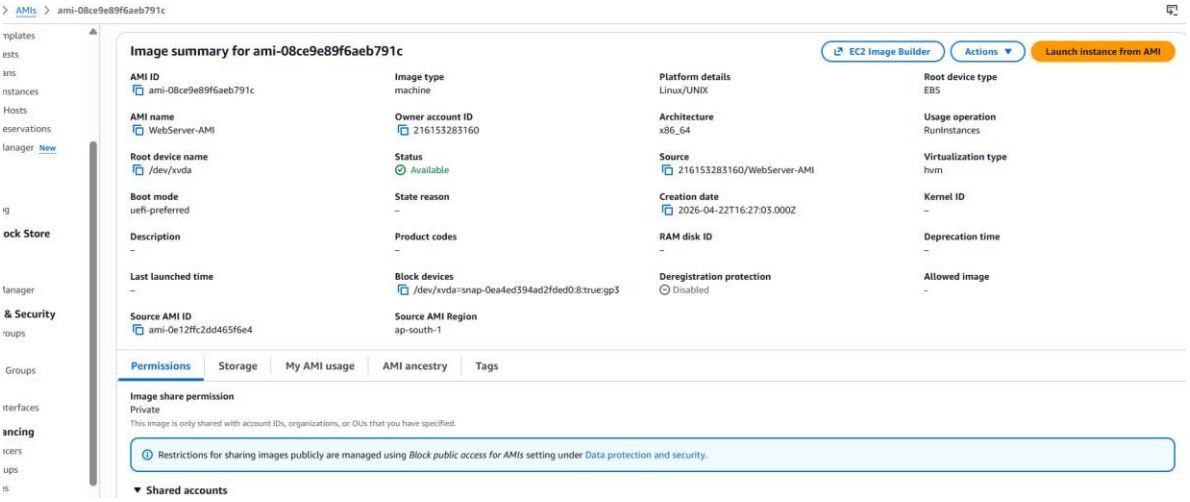
ScaleX–Scalable AWS web infrastructure with auto scaling

Step1: Create EC2 Instance: (webserver-1)



Step2: Create AMI

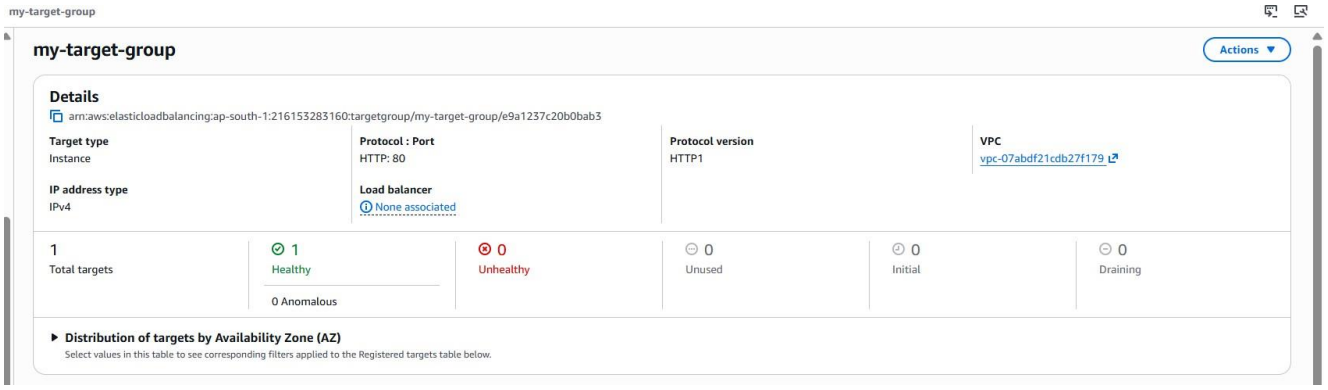




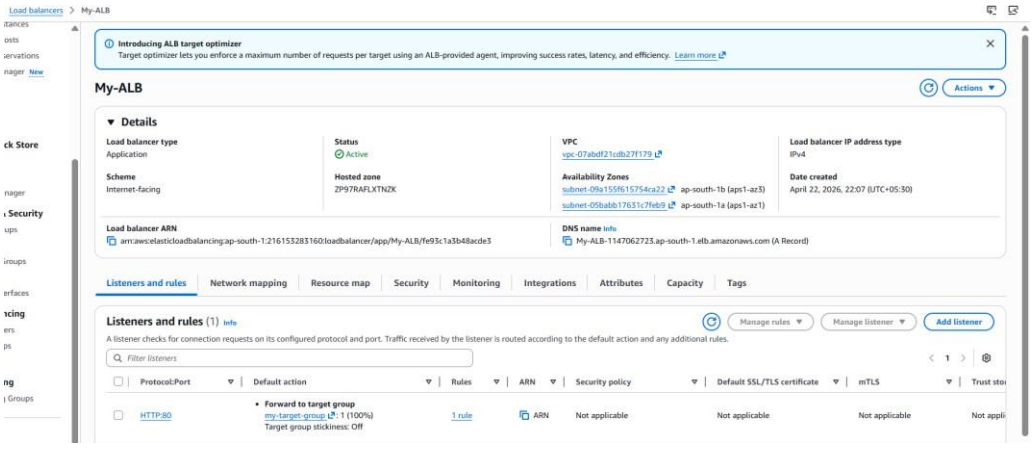
Step 3: Connect with SSH and enter a command:

```
  Verifying        : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64                                                    4/13
  Verifying        : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch                                               5/13
  Verifying        : httpd-2.4.66-1.amzn2023.0.1.x86_64                                                              6/13
  Verifying        : httpd-core-2.4.66-1.amzn2023.0.1.x86_64                                                         7/13
  Verifying        : httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch                                                   8/13
  Verifying        : httpd-tools-2.4.66-1.amzn2023.0.1.x86_64                                                        9/13
  Verifying        : libbrotli-1.0.9-4.amzn2023.0.2.x86_64                                                          10/13
  Verifying        : mailcap-2.1.49-3.amzn2023.0.3.noarch                                                           11/13
  Verifying        : mod_http2-2.0.27-1.amzn2023.0.3.x86_64                                                         12/13
  Verifying        : mod_lua-2.4.66-1.amzn2023.0.1.x86_64                                                           13/13

Installed:
  apr-1.7.5-1.amzn2023.0.4.x86_64                    apr-util-1.6.3-1.amzn2023.0.2.x86_64                    apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64
  apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64       generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch       httpd-2.4.66-1.amzn2023.0.1.x86_64
  httpd-core-2.4.66-1.amzn2023.0.1.x86_64            httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch           httpd-tools-2.4.66-1.amzn2023.0.1.x86_64
  libbrotli-1.0.9-4.amzn2023.0.2.x86_64              mailcap-2.1.49-3.amzn2023.0.3.noarch                    mod_http2-2.0.27-1.amzn2023.0.3.x86_64
  mod_lua-2.4.66-1.amzn2023.0.1.x86_64

Complete!
[ec2-user@ip-172-31-34-117 ~]$ sudo systemctl start httpd
[ec2-user@ip-172-31-34-117 ~]$ sudo systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
```

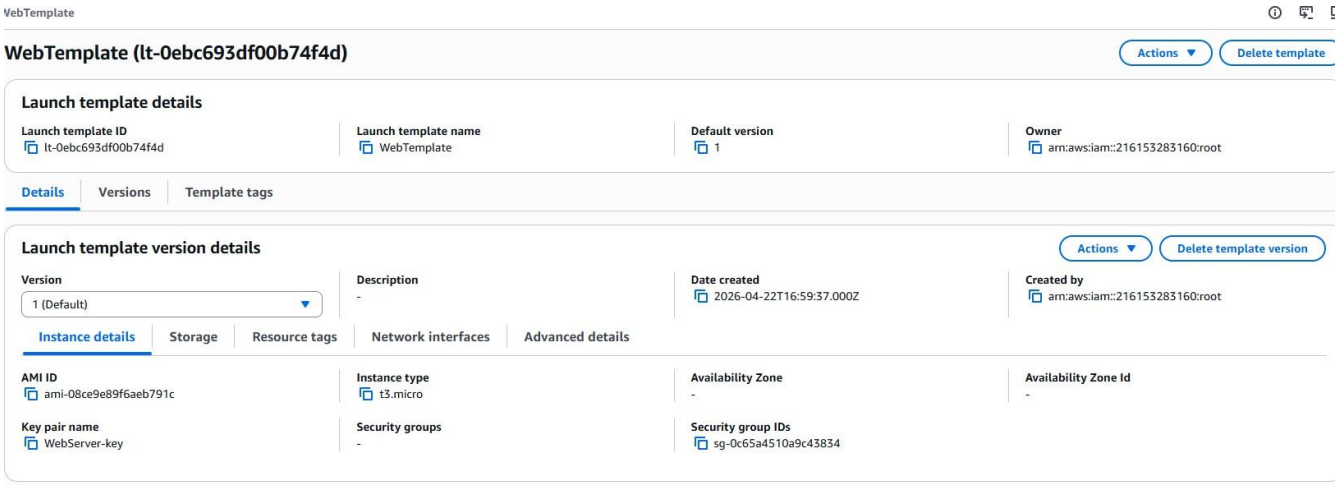
Step 4: Create Target Group



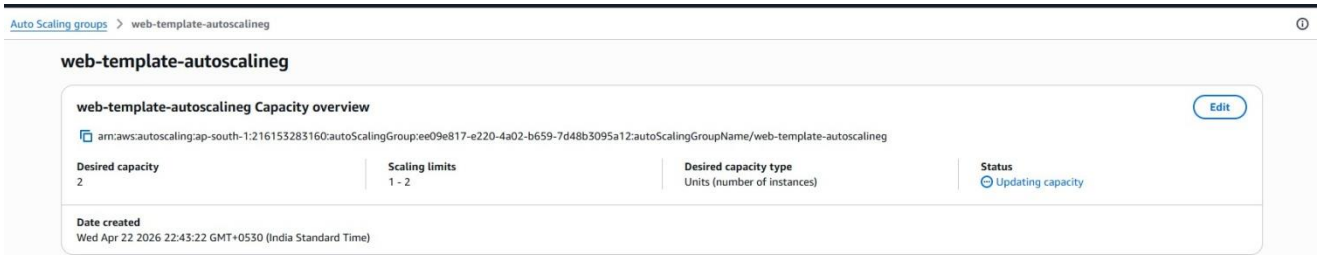
Step 5: Create Load Balancer



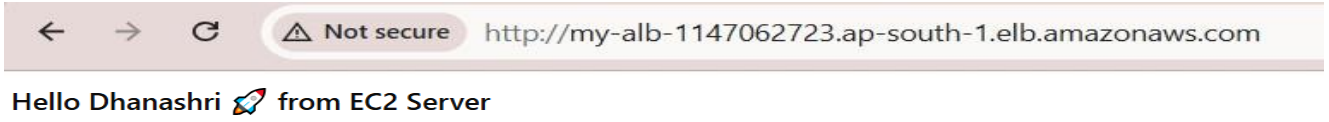
Step 6: Create Launch Template



Step 7: Create Auto Scaling (SNS Enable for Email Notification)



Final Output:



This project successfully implements a scalable and highly available web application architecture on AWS using EC2, Application Load Balancer, and Auto Scaling. The system distributes incoming traffic efficiently across multiple instances, automatically adjusts resources based on demand, and ensures minimal downtime through health checks and failover mechanisms. Additionally, SNS integration provides real-time notifications for instance launch and termination events, making the system reliable, automated, and production-ready.

SNS Notification:

AWS Notifications <no-reply@sns.amazonaws.com> 23:06 (27 minutes ago)

to me

Service: AWS Auto Scaling
Time: 2026-04-22T17:36:31.380Z
RequestId: acf676d8-fce0-52b2-1363-701ddd56f39a
Event: autoscaling:EC2_INSTANCE_TERMINATE
AccountId: 216153283160
AutoScalingGroupName: web-template-autoscalineg
AutoScalingGroupARN: arn:aws:autoscaling:ap-south-1:216153283160:autoScalingGroup:ee09e817-e220-4a02-b659-7d48b3095a12:autoScalingGroupName/web-template-autoscalineg
ActivityId: acf676d8-fce0-52b2-1363-701ddd56f39a
Description: Terminating EC2 instance: i-095ddcc8319b5a944
Cause: At 2026-04-22T17:31:29Z an instance was taken out of service in response to an EC2 health check indicating it has been terminated or stopped.
StartTime: 2026-04-22T17:31:29.684Z
EndTime: 2026-04-22T17:36:31.380Z
StatusCode: InProgress
StatusMessage:
Progress: 50
EC2InstanceId: i-095ddcc8319b5a944

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